

SQL Window Functions & Advanced Analytics

20 Interview Questions — Northwind Database

The Complete DSA + SQL Interview Preparation Series

Book 6 of 13

For placement preparation, coding interviews, and product-based company rounds.

About This Book

This is the third and final SQL book in the series, and it completes the full 50-question SQL set (10 from Book 4, 20 from Book 5, 20 here). It covers the questions that separate a comfortable SQL user from someone genuinely interview-ready: window functions like RANK, ROW_NUMBER, and running totals, common table expressions, and multi-step real-world business analytics questions that combine several techniques in one query.

These are the questions most likely to appear in senior or product-company SQL rounds, because they test whether you can decompose an ambiguous business ask into a precise, efficient query — not just recall syntax.

How To Use This Book

Before reaching for a window function, ask: am I ranking, comparing to a previous row, or computing a running/rolling value? That question alone points you to RANK/ROW_NUMBER, LAG/LEAD, or a windowed SUM/AVG respectively. For the real-world scenario questions, write out the business logic in plain English first — the SQL is usually the easy part once the logic is clear.

Difficulty Breakdown

Difficulty	Count
Medium	9
Hard	11

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Q1. Rank Customers by Total Spend

Medium

Topic: RANK() Window Function

Business Scenario

The account management team wants every customer ranked by total lifetime spend, with ties handled properly (two customers with identical spend should share the same rank).

TABLES USED

Customers, Orders, OrderDetails

EXPECTED OUTPUT COLUMNS

CompanyName, TotalSpend, SpendRank

Sample Result (illustrative)

CompanyName	TotalSpend	SpendRank
QUICK-Stop	117483.39	1
Ernst Handel	104874.98	2

Skills Required

RANK()	OVER()	JOIN	SUM()
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Interview Tip

Window functions never collapse rows the way GROUP BY does — you get one output row per input row, with the aggregate calculated 'alongside' it. That distinction trips up a lot of candidates.

Q2. Number Each Order Within a Customer's History

Medium

Topic: ROW_NUMBER()

Business Scenario

A customer timeline feature needs each customer's orders numbered sequentially (1st order, 2nd order, and so on) in the order they were placed, resetting the count for every customer.

TABLES USED

Orders

EXPECTED OUTPUT COLUMNS

CustomerID, OrderID, OrderDate, OrderSequence

Sample Result (illustrative)

CustomerID	OrderID	OrderDate	OrderSequence
ALFKI	10643	1997-08-25	1
ALFKI	10692	1997-10-03	2

Skills Required

ROW_NUMBER()	PARTITION BY	ORDER BY
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Interview Tip

PARTITION BY is the window-function equivalent of GROUP BY: it resets the calculation for each group without removing any rows.

Q3. Running Total of Daily Revenue

Hard

Topic: Window SUM (Running Total)

Business Scenario

Finance wants a day-by-day running cumulative revenue total across the whole year, so they can plot progress toward the annual target as it accumulates.

TABLES USED

Orders, OrderDetails

EXPECTED OUTPUT COLUMNS

OrderDate, DailyRevenue, RunningTotal

Sample Result (illustrative)

OrderDate	DailyRevenue	RunningTotal
1996-07-04	440.00	440.00
1996-07-05	1863.40	2303.40

Skills Required

Window SUM()

ORDER BY

GROUP BY

Interview Tip

For running totals, the ORDER BY inside the OVER() clause determines the accumulation order — get this wrong and the running total will still 'work', just incorrectly.

Q4. Each Order's Value vs. the Customer's Previous Order

Hard

Topic: LAG()

Business Scenario

A churn-risk model wants to compare each order's value to that same customer's immediately preceding order, to flag customers whose spend is trending down order over order.

TABLES USED

Orders, OrderDetails

EXPECTED OUTPUT COLUMNS

CustomerID, OrderID, OrderValue, PreviousOrderValue

Sample Result (illustrative)

CustomerID	OrderID	OrderValue	PreviousOrderValue
ALFKI	10692	878.00	814.50
ALFKI	10702	330.00	878.00

Skills Required

LAG()

PARTITION BY

OVER()

Interview Tip

LAG() and LEAD() need an explicit ORDER BY inside OVER() to mean anything; without it, 'previous row' is undefined.

Q5. Top 2 Products by Revenue Within Each Category

Hard

Topic: Window RANK per Group

Business Scenario

Merchandising wants the two top-selling products by revenue within every category, not just overall, so each category gets fair representation in a 'best sellers' homepage widget.

TABLES USED

Categories, Products, OrderDetails

EXPECTED OUTPUT COLUMNS

CategoryName, ProductName, Revenue, RankInCategory

Sample Result (illustrative)

CategoryName	ProductName	Revenue	RankInCategory
Beverages	Côte de Blaye	141396.74	1
Beverages	Ipoh Coffee	25079.20	2

Skills Required

RANK()	PARTITION BY	JOIN
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Interview Tip

When ranking within groups (e.g., top N per category), PARTITION BY the group and ORDER BY the ranking metric, then filter the rank in an outer query — RANK() can't be filtered directly in the same SELECT.

Q6. Percentage of Total Revenue by Category

Hard

Topic: Window SUM as Denominator

Business Scenario

Leadership wants each category's revenue expressed as a percentage of total company revenue, to visualize category mix in a single pie chart.

TABLES USED

Categories, Products, OrderDetails

EXPECTED OUTPUT COLUMNS

CategoryName, CategoryRevenue, PercentOfTotal

Sample Result (illustrative)

CategoryName	CategoryRevenue	PercentOfTotal
Beverages	286526.95	19.8%
Dairy Products	251330.50	17.4%

Skills Required

Window SUM()

Aggregate Ratios

GROUP BY

Interview Tip

Percentage-of-total questions usually need a window SUM() with no PARTITION BY (or partitioned only by the top-level group) as the denominator, alongside a grouped or windowed numerator.

Q7. Customers With a Common Table Expression for Repeat Buyers

Medium

Topic: CTE

Business Scenario

Marketing wants a cleanly readable query (using a CTE rather than a nested subquery) that returns customers who placed more than 3 orders in the last twelve months of available data.

TABLES USED

Customers, Orders

EXPECTED OUTPUT COLUMNS

CompanyName, OrderCountLastYear

Sample Result (illustrative)

CompanyName	OrderCountLastYear
Save-a-lot Markets	9
Hungry Owl All-Night Grocers	7

Skills Required

WITH (CTE)	GROUP BY	HAVING
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Interview Tip

CTEs don't change performance by themselves — their real value in an interview is proving you can decompose a complex question into readable, named steps.

Q8. Month-over-Month Revenue Growth

Hard

Topic: LAG() / Percentage Change

Business Scenario

Leadership wants to see month-over-month revenue growth as a percentage, comparing each month's total revenue to the previous month's, to spot acceleration or slowdown.

TABLES USED

Orders, OrderDetails

EXPECTED OUTPUT COLUMNS

OrderMonth, Revenue,
PriorMonthRevenue, GrowthPercent

Sample Result (illustrative)

OrderMonth	Revenue	PriorMonthRevenue	GrowthPercent
1996-08	25485.28	27861.90	-8.5%
1996-09	26381.40	25485.28	3.5%

Skills Required

LAG()

Date Functions

Calculated Columns

Interview Tip

Growth/change-over-time questions almost always combine LAG() with a calculated percentage column — practice this pattern until it's automatic.

Q9. Median Order Value per Employee

Hard

Topic: Percentile / Window Function

Business Scenario

Sales operations wants the median order value handled by each employee (not the average, which can be skewed by a few very large orders), to more fairly compare performance.

TABLES USED

Employees, Orders, OrderDetails

EXPECTED OUTPUT COLUMNS

FirstName, LastName, MedianOrderValue

Sample Result (illustrative)

FirstName	LastName	MedianOrderValue
Nancy	Davolio	642.10
Andrew	Fuller	701.35

Skills Required

PERCENTILE_CONT()	Window Functions	GROUP BY
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Interview Tip

True median requires PERCENTILE_CONT or a manual row-counting trick — AVG() is not a substitute, and interviewers will probe whether you know the difference.

Q10. First and Last Order Date per Customer

Medium

Topic: MIN / MAX / GROUP BY

Business Scenario

Customer success wants each customer's first and most recent order date side by side, to calculate how long each account has been active.

TABLES USED

Customers, Orders

EXPECTED OUTPUT COLUMNS

CompanyName, FirstOrderDate, LastOrderDate

Sample Result (illustrative)

CompanyName	FirstOrderDate	LastOrderDate
Alfreds Futterkiste	1997-08-25	1998-04-09
Ana Trujillo Emparedados	1996-09-18	1998-03-04

Skills Required

MIN()	MAX()	GROUP BY
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Interview Tip

MIN/MAX-per-group questions are often disguised 'first/last event' questions — recognize the pattern regardless of the business wording.

Q11. Customers Who Bought From Every Category

Hard

Topic: Subquery / Set Logic

Business Scenario

Merchandising wants to identify the small set of customers who have purchased at least one product from every single product category, as candidates for a cross-category loyalty reward.

TABLES USED

Customers, Orders, OrderDetails,
Products, Categories

EXPECTED OUTPUT COLUMNS

CompanyName

Sample Result (illustrative)

CompanyName
Rattlesnake Canyon Grocery

Skills Required

Subquery	GROUP BY	HAVING COUNT(DISTINCT)
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Interview Tip

'Bought from every category' is a classic set-containment question — mentally translate it to 'distinct category count per customer equals total category count'.

Q12. Employee Sales Compared to the Team Average

Hard

Topic: Window AVG()

Business Scenario

Sales leadership wants each employee's total sales shown next to the overall team average, and the difference between them, in the same result set — no separate query for the average.

TABLES USED

Employees, Orders, OrderDetails

EXPECTED OUTPUT COLUMNS

FirstName, LastName, EmployeeSales, TeamAverage, Difference

Sample Result (illustrative)

FirstName	LastName	EmployeeSales	TeamAverage	Difference
Margaret	Peacock	232890.85	185184.36	47706.49
Michael	Suyama	78198.10	185184.36	-106986.26

Skills Required

Window AVG()	OVER()	Calculated Columns
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Interview Tip

Comparing an individual to a group average in the same row is the signature use case for a window AVG() — no self-join or subquery required.

Q13. Products That Are Best Sellers in More Than One Category's Ranking

Hard

Topic: CTE + RANK()

Business Scenario

A cross-category analysis needs any product that ranks in the top 3 by revenue within its category AND is also in the overall top 20 products company-wide, to see which local winners are also global winners.

TABLES USED

Products, Categories, OrderDetails

EXPECTED OUTPUT COLUMNS

ProductName, CategoryName, CategoryRank, OverallRank

Sample Result (illustrative)

ProductName	CategoryName	CategoryRank	OverallRank
Côte de Blaye	Beverages	1	1

Skills Required

CTE	RANK()	PARTITION BY	JOIN
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Interview Tip

Multi-level ranking questions (top per group AND top overall) usually need two ranking passes, often via a CTE that first computes per-group rank.

Q14. Orders Placed on Weekends

Medium

Topic: Date Functions

Business Scenario

Operations is exploring whether to adjust weekend staffing and wants a count of how many orders were placed on Saturday or Sunday versus weekdays.

TABLES USED

Orders

EXPECTED OUTPUT COLUMNS

DayType, OrderCount

Sample Result (illustrative)

DayType	OrderCount
Weekday	802
Weekend	28

Skills Required

Date Functions	CASE	GROUP BY
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Interview Tip

Weekday/weekend or time-bucket questions are CASE expressions wrapped around a date function — keep the CASE logic simple and readable.

Q15. Customer Cohort: First Purchase Month

Hard

Topic: CTE / Cohort Analysis

Business Scenario

Growth wants each customer tagged with the month of their very first order (their 'cohort'), as the foundation for a retention curve analysis in a later step.

TABLES USED

Customers, Orders

EXPECTED OUTPUT COLUMNS

CompanyName, CohortMonth

Sample Result (illustrative)

CompanyName	CohortMonth
Alfreds Futterkiste	1997-08
Ana Trujillo Emparedados	1996-09

Skills Required

CTE

MIN()

Window Functions

Interview Tip

Cohort analysis always starts with 'first event date per entity' — get comfortable computing that with MIN() before adding any complexity on top.

Q16. Products With Declining Order Frequency

Hard

Topic: LAG() / Trend Detection

Business Scenario

Inventory planning wants to flag products whose order count has decreased in each of the last two consecutive quarters, as early candidates for discontinuation.

TABLES USED

Products, OrderDetails, Orders

EXPECTED OUTPUT COLUMNS

ProductName, Q1Count, Q2Count, Q3Count

Sample Result (illustrative)

ProductName	Q1Count	Q2Count	Q3Count
Chef Anton's Gumbo Mix	18	12	7

Skills Required

LAG()	Date Functions	CASE
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Interview Tip

Trend-decline questions over multiple periods often need several LAG() calls chained, or a self-join across period boundaries — plan the shape before writing SQL.

Q17. Employees Whose Sales Beat Their Manager's Average Team

Hard

Topic: Correlated Subquery + Self Join

Business Scenario

HR wants to find employees whose individual total sales exceed the average total sales of everyone reporting to the same manager, as a signal for standout performer recognition.

TABLES USED

Employees, Orders, OrderDetails

EXPECTED OUTPUT COLUMNS

EmployeeName, ManagerName, EmployeeSales, ManagerTeamAverage

Sample Result (illustrative)

EmployeeName	ManagerName	EmployeeSales	ManagerTeamAverage
Janet Leverling	Andrew Fuller	202812.75	150331.20

Skills Required

Self Join	Correlated Subquery	AVG()
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Interview Tip

'Compare to peers under the same manager' is a hint to self-join the employee table and then aggregate on the manager side.

Q18. Days Between Consecutive Orders per Customer

Hard

Topic: LAG() / Date Difference

Business Scenario

A retention model needs the number of days between each customer's orders and their immediately previous order, to estimate a typical reorder cycle per customer.

TABLES USED

Orders

EXPECTED OUTPUT COLUMNS

CustomerID, OrderDate, PreviousOrderDate, DaysBetween

Sample Result (illustrative)

CustomerID	OrderDate	PreviousOrderDate	DaysBetween
ALFKI	1997-10-03	1997-08-25	39
ALFKI	1998-03-24	1997-10-03	172

Skills Required

LAG()	Date Math	PARTITION BY
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Interview Tip

Any 'days since' or 'days between' question is date subtraction paired with LAG() when comparing to a previous row, or NOW()/CURRENT_DATE when comparing to today.

Q19. Top Supplier by Revenue in Each Country

Hard

Topic: Window RANK per Group

Business Scenario

Procurement strategy wants the single highest-revenue-generating supplier headquartered in each country, to understand geographic supplier concentration by revenue rather than just headcount.

TABLES USED

Suppliers, Products, OrderDetails

EXPECTED OUTPUT COLUMNS

Country, CompanyName, SupplierRevenue

Sample Result (illustrative)

Country	CompanyName	SupplierRevenue
USA	New England Seafood Cannery	48267.20
UK	Specialty Biscuits, Ltd.	31655.90

Skills Required

RANK()	PARTITION BY	JOIN
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Interview Tip

Geographic 'top X per region' questions follow the exact same shape as 'top X per category' — PARTITION BY the region, RANK by the metric.

Q20. Full Customer Health Snapshot

Medium

Topic: Multi-CTE Business Report

Business Scenario

The exec team wants a single report combining, per customer: total orders, total spend, average order value, and days since their last order — the standard 'account health' view used in quarterly reviews.

TABLES USED

Customers, Orders, OrderDetails

EXPECTED OUTPUT COLUMNS

CompanyName, TotalOrders, TotalSpend, AvgOrderValue, DaysSinceLastOrder

Sample Result (illustrative)

CompanyName	TotalOrders	TotalSpend	AvgOrderValue	DaysSinceLastOrder
QUICK-Stop	28	117483.39	4195.12	45
Ernst Handel	30	104874.98	3495.83	12

Skills Required

CTE	JOIN	Multiple Aggregates	Date Math
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Interview Tip

Full 'account health' or 'customer 360' style reports are really several smaller questions stitched together — build and test each metric separately before combining them.